FAIR DATA INNOVATIONS HUB

CALIFORNIA MEDICAL INNOVATIONS INSTITUTE

Toward the Envision Portal: Designing a FAIR and AI-Ready Framework for Ophthalmic Imaging Sharing and Discovery

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Background

Ophthalmic imaging datasets, such as optical coherence tomography (OCT) and fundus photographs, are a valuable resource for advancing AI/ML models for disease detection.

However, they are **not consistently shared**. When shared, their reuse is hindered by fragmented sharing that limits findability and by inconsistent structure that reduces interoperability and machine usability (AI/ML).

Purpose

We are developing the **Envision Portal (EP)**, an open source platform for easily sharing, finding, and reusing ophthalmic imaging data. We are currently focusing on two major features:

- **Automation** to format ophthalmic imaging data into FAIR (Findable, Accessible, Interoperable, Reusable) and machine-usable datasets
- Pipeline to **discover** ophthalmic imaging datasets from data repositories and register them in the EP database to create the largest registry of ophthalmic datasets

Methods

1. We reviewed FAIR data principles and ophthalmic imaging standards, including guidelines from NIH Bridge2AI.
2. We developed a SetFit few-shot classifier trained to classify dataset metadata (title, description, keywords, and file types) as ophthalmic imaging dataset or not. The classifier is built on the all-mpnet-base-v2 sentence transformer (768-dim) and was trained on 891 curated examples (262 positive, 629 negative).

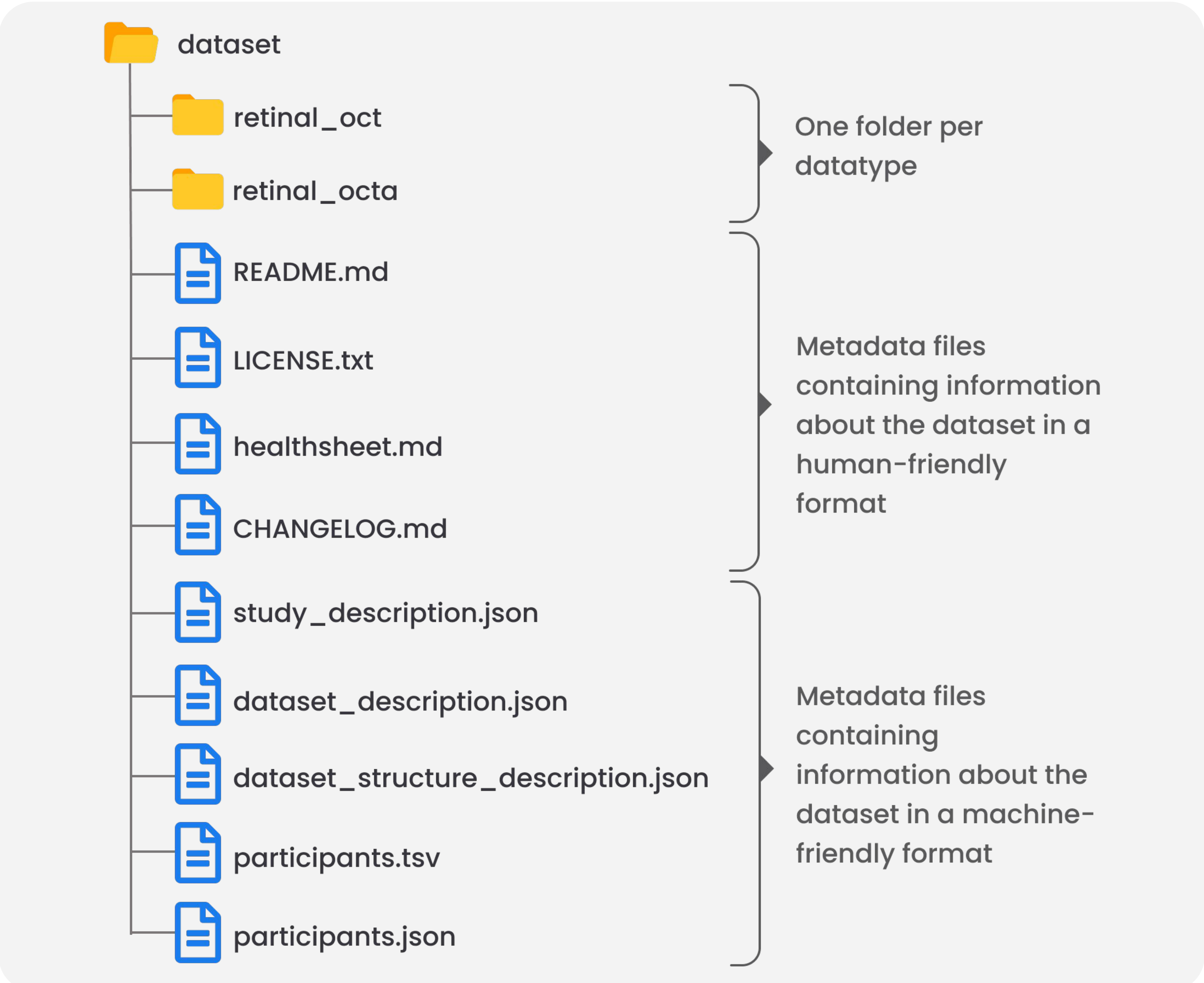

Figure 1. Illustration of a dataset organized according to the Clinical Dataset Structure (CDS) developed by the AI-READI project in the NIH Bridge2AI program.

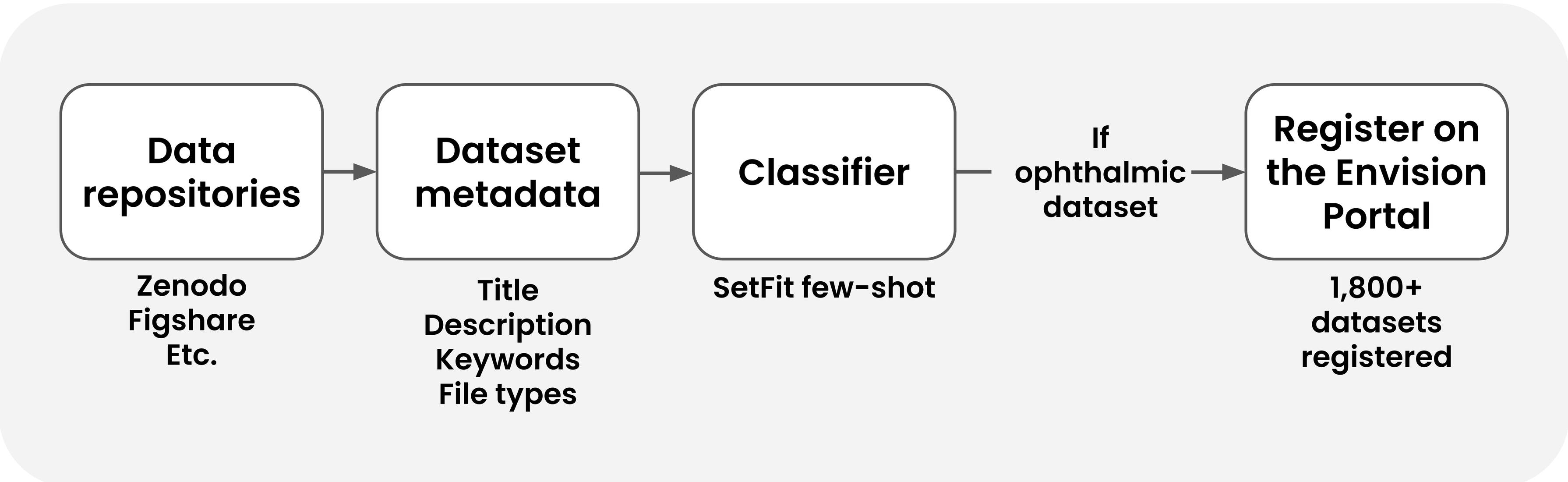

Figure 2. Illustration of the structure of the pipeline to identify ophthalmic datasets based on their metadata.

Results

We finalized major design decisions and established the base framework of the Envision Portal:

- The Clinical Dataset Structure (CDS) will be used for creating standard format datasets (Figure 1).
- DICOM will be used as the preferred format for imaging data.
- The SetFit classifier achieved a test accuracy of 0.961 and and F1 of 0.936 on the positive class. Expert validation is currently ongoing.
- Applied on datasets from six repositories, the pipeline (Figure 2) identified 1,800+ ophthalmic imaging datasets that can be browsed on the platform at envisionportal.org.

Conclusion

Reviewing existing imaging standards allowed us to identify the key components a standardized ophthalmic imaging dataset should contain, leading to the adoption of CDS and DICOM as the foundation for the Envision Portal.

Our SetFit-based classifier demonstrates that cost-effective ophthalmic imaging dataset discovery is achievable from metadata alone, without accessing underlying image files. This will be further validated through an expert curated training set.

Future work will focus on expanding repository coverage for dataset discovery and developing tools to format any set of ophthalmic data files into our prescribed dataset format.

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envisionportal.org
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